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### Q69. Maximum Sum Rectangle in Matrix

Given a 2D matrix of integers, find the sub-rectangle with the maximum sum. The rectangle must be contiguous. The objective is to maximize the sum. Use Dynamic Programming with Kadane's approach.

Input Format:

matrix[][]

Sample Input:

matrix =

1 2 -1

-3 4 5

Expected Output:

Max Sum = 9

Ans:

```
def kadane(arr):
```

```
    max_sum = arr[0]
```

```
    current_sum = arr[0]
```

```
    for i in range(1, len(arr)):
```

```
        current_sum = max(arr[i], current_sum + arr[i])
```

```
        max_sum = max(max_sum, current_sum)
```

```
    return max_sum
```

```
def maximum_sum_rectangle(matrix):
```

```
    rows = len(matrix)
```

```
    cols = len(matrix[0])
```

```
max_sum = float("-inf")
```

```
for left in range(cols):
```

```
    temp = [0] * rows
```

```
    for right in range(left, cols):
```

```
        for i in range(rows):
```

```
            temp[i] += matrix[i][right]
```

```
        current_sum = kadane(temp)
```

```
        max_sum = max(max_sum, current_sum)
```

```
    return max_sum
```

```
matrix = [  
    [1, 2, -1],  
    [-3, 4, 5]  
]
```

```
answer = maximum_sum_rectangle(matrix)
```

```
print("Max Sum =", answer)
```

### Q70. Minimum Insertions to Balance Parentheses

Given a string of parentheses, determine the minimum number of insertions required to make it valid. The objective is to balance the parentheses. Efficient counting is required. Use Greedy or Dynamic Programming.

Input Format:

s = parentheses string

Sample Input:

s = (()

Expected Output:

Min Insertions = 1

Ans:

```
def minimum_insertions_to_balance(s):
```

```
    insertions = 0
```

```
    open_count = 0
```

```
    for ch in s:
```

```
        if ch == "(":
```

```
            open_count += 1
```

```
        elif ch == ")":
```

```
            if open_count > 0:
```

```
                open_count -= 1
```

```
            else:
```

```
                insertions += 1
```

```
    insertions += open_count
```

```
    return insertions
```

```
s = "()"
```

```
answer = minimum_insertions_to_balance(s)
```

```
print("Min Insertions =", answer)
```